

Correlated order flow: Pervasiveness, sources, and pricing effects

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Abstract

We hypothesize that a combination of indexing, industry and broader market forces create common effects in order flow and returns. We test the relative contribution of each to common effects in large samples of both index and non-index stocks. Common effects are strong in index constituent stocks, but are economically inconsequential in non-index stocks. Once indexing effects have been removed, the common effects in index constituents disappear. Industry and broader market effects exist, but contribute relatively little to common effects. Aside from identifying economic causes for the statistical common factors extracted in prior work, these results indicate that common effects are not pervasive. Finally, we show that common effects in order flow have real economic impact by linking them to common effects in returns. This effect of order flow on the correlation structure of returns has implications for diversification strategies. The results also show that specialists adjust prices differentially in the presence of aggregate order flow, supporting multi-asset models of price formation and suggesting that traders with discretion can minimize the price impact of their trades by timing these to coincide with index trading.

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Abstract

We hypothesize that a combination of indexing, industry and broader market forces create common effects in order flow and returns. We test the relative contribution of each to common effects in large samples of both index and non-index stocks. Common effects are strong in index constituent stocks, but are economically inconsequential in non-index stocks. Once indexing effects have been removed, the common effects in index constituents disappear. Industry and broader market effects exist, but contribute relatively little to common effects. Aside from identifying economic causes for the statistical common factors extracted in prior work, these results indicate that common effects are not pervasive. Finally, we show that common effects in order flow have real economic impact by linking them to common effects in returns. This effect of order flow on the correlation structure of returns has implications for diversification strategies. The results also show that specialists adjust prices differentially in the presence of aggregate order flow, supporting multi-asset models of price formation and suggesting that traders with discretion can minimize the price impact of their trades by timing these to coincide with index trading.

1 Introduction

Recent research has uncovered common effects in trading activity, returns and trading costs for individual stocks. Hasbrouck and Seppi (2001) use factor analysis to reveal comovement in intraday order flow, returns and trading costs across the 30 Dow Jones constituents, and canonical correlations to establish that the common factors are related.¹ Chordia, Roll and Subrahmanyam (2000) document the existence of market and industry factors in average daily changes in five measures of liquidity across a broad sample of NYSE stocks (see, also, Huberman and Halka, 2001).² Common effects in longer-horizon trading volume are identified by Tkac (1999), via market model type regressions, and Lo and Wang (2000), via factor analysis.

Our study extends this literature by exploring the following unanswered questions about common effects or *commonality* (following Chordia, Roll and Subrahmanyam) in order flow and returns. First, how pervasive are common effects in order flow and returns, and what are their sources? Because the extant evidence on commonality is based on samples dominated by stocks in major indexes, we hypothesize that trading by index funds may be the major source of commonality. This implies that samples of non-index stocks will exhibit little in the way of common effects. Second, are commonality and its effects getting stronger through time? If indexing is at the root of common effects, then these effects should be growing as indexing grows. Finally, is commonality in returns actually produced by commonality in order flow? Unless commonality in order flow affects returns, it is of little economic importance.

Our approach is to hypothesize a hierarchy of non-mutually exclusive sources of common effects, comprising industry, index and broader-market factors. Using intraday data, we study common effects for stocks in the S&P 500 index and a large sample of non-S&P stocks for both 1986 and 1996. In each year, we examine both the end of the day (when index funds are most active) and the rest of the day.

¹ Order flow is the difference between buy and sell volume or number of transactions. Because it indicates the net direction of trades, it is sometimes referred to as net order flow.

² The existence of common factors in lower frequency (daily, weekly or monthly) returns and even volume has been known for as long as 30 years, as evident in the use of the market model or its variants.

Our tests proceed as follows. First, we establish a baseline level of order flow commonality in the two samples in both 1986 and 1996 by calculating the average correlation in intraday (fifteen-minute) order flow for all pairs of stocks in the S&P and non-S&P samples, and do the same for intraday returns. We then build models that progressively add sources of common effects (industry, index and broader market order flow and returns) to explain an individual stock's order flow. By examining the increase in R-squared at each stage and the decrease in the pairwise correlations of the residual order flows, we can measure the contribution of each hypothesized source of commonality. In the second stage of our analysis, we estimate models that link an individual stock's returns to order flow coming from the hypothesized sources of commonality.

We find strong common effects in order flow for index stocks and find that once indexing effects are removed, the commonality drops to almost nil. The incremental effects of industry or broader market factors on order flow for index constituents, while statistically significant, are economically negligible. Consistent with the fact that indexing is driving common effects, we find that commonality in order flow increases over time in step with the interest in indexing. Further, common effects are generally strongest at the end of the day, when index funds are most active. We broaden the search for common effects to stocks that are not in the S&P 500 index. The results show that non-index stocks have statistically discernible, but economically small, common effects, and that their order flow bears an association with industry and broader market effects.

Next, we link common effects in order flow to common effects in intraday returns. Again, the commonality in returns is primarily confined to index stocks and exhibits the same characteristics of growth over time and relative strength at the end of the day. Once common index order flow is accounted for, return commonality drops by 50 to 70%. The returns of index constituents exhibit a slightly weaker sensitivity to industry order flow. Returns for non-index stocks display sensitivity to broader market order flow and the return correlation declines appreciably once broader market or industry order flow is incorporated.

Finally, we estimate principal components for order flow and returns in our two samples. Our objective is to examine the correspondence, if any, between statistically extracted factors and our economic factors. In each sample, while several principal components are statistically significant, the first principal component is dominant in economic terms, explaining 10% of order flow variation for index stocks, and slightly less for non-index stocks. The first principal component for S&P stocks is highly correlated with aggregate S&P order flow, while that for non-S&P stocks is highly correlated with aggregate non-S&P order flow. The principal components for returns explain slightly less of the overall variation, though there is evidence that the first (and dominant) principal component is correlated with aggregate S&P and non-S&P order flow in the two samples. Thus, we find correspondence between our hypothesized economic variables and the statistically extracted factors.

Establishing indexing as the primary source of common effects offers several insights. First, it shows that the common order flow effects identified by Hasbrouck and Seppi (2001) in a sample of exclusively index stocks are not pervasive in other stocks. Nonetheless, even though common effects are economically large only for constituents of the S&P 500, these stocks represent approximately two-thirds of U.S. market capitalization. Second, our results clearly identify tangible effects of indexing on the trading environment of stocks in the popular indexes. Third, the common effects in order flow induced by indexing are directly linked to commonality in returns. Thus, these common effects in order flow have an economic impact.

More generally, our results shed light on the potentially damaging effects of common trading on attempts to diversify, and on holding index funds. Barberis, Shleifer and Wurgler (2001) provide an explanation for return comovement based on correlated uninformed trading by investors interested in particular groups of assets. Consistent with correlated demand on the part of clienteles (e.g. index funds), they document a higher beta with respect to the index and a larger market model R-squared after a stock is added to the S&P 500. Our findings that index constituents display stronger intraday order flow and return covariation than do non-index stocks, and this greater return covariation can

be attributed to the greater order flow covariation, are consistent with their trading-based explanation.

Since common effects in trading activity affect the correlation structure of returns, they simultaneously affect the variance of diversified portfolios. Our results raise the possibility that, by increasing return correlations, common order flow is undermining the effectiveness of portfolio diversification strategies. Moreover, if common effects are limited to constituents of major indexes, it might be possible for an investor to mitigate the pricing effects associated with the growing importance of index funds by holding a broader portfolio consisting of non-index stocks. (See, also Hasbrouck and Seppi, 2001, and Chordia et al., 2000, 2001).

Finally, the incremental explanatory power of aggregate order flow for returns has implications for theoretical models of specialist behavior. Specifically, this result suggests that the specialist learns from order flow in other stocks, and supports multi-asset models of price formation such as Caballe and Krishnan (1994) and others. To our knowledge, the only other study of basket trading effects on individual stocks is that of Hasbrouck (1996) who, using TORQ data, finds that program and non-program trades have similar price impacts.

The rest of the paper is organized as follows. Section 2 describes our hypotheses. Section 3 summarizes our data and methods, and section 4 contains our results. Section 5 discusses the implications of these results, and section 6 concludes.

2 Hypothesis development

The sources of order flow commonality. In attempting to identify the factors driving order flow in stocks, we start with three hypotheses concerning the sources of common order flow. These sources represent a non-mutually exclusive hierarchy of influences: broad market effects, indexing effects, and industry effects. We will assess the relative importance of each hypothesized source in explaining commonalities in order flow.

The first hypothesized source of commonality is broad shifts in investor allocations, taking the form of increased flows to equity mutual funds, or asset allocation shifts from

bonds to stocks, for example. Edelen (1999) shows that monthly flows to equity funds are significantly correlated—the average correlation between individual fund flows and aggregate market flows is +0.35. Likewise, Edelen and Warner (2001) show that aggregate daily flows have systematic effects, which suggests that there is a common component to the flows across funds. Thus, the *broad market effects* hypothesis predicts the existence of commonality in order flow for all stocks, regardless of industry or index membership.

The second source of common effects is correlated industry events. Stocks in the same industry or in related industries will exhibit common order flow due to common information, or industry or sector mutual funds. Thus, the *industry effects* hypothesis predicts a strong common order flow factor for firms in the same industry. Note that an industry factor can explain common effects within an industry but not pervasive common effects across industries.

The third hypothesized source of commonality is indexing. The dramatic growth in indexing, both by mutual funds and by pension funds, presents the possibility that the constituents of popular indexes are subject to strong common order flow effects. Since the S&P 500 is the index most widely followed by index funds and also by index arbitrageurs, firms in the S&P 500 index face a clearly delineated source of commonality.³ The *indexing* hypothesis predicts that order flow for S&P stocks will exhibit substantial common effects, and will be associated mainly with aggregate S&P order flow and returns.

By contrast, the broad market effects hypothesis predicts that there also will be strong common effects in order flow for non-S&P stocks. Additionally, the broad market effects and industry effects hypotheses predict that such factors as non-S&P and industry order flow and returns will play an important part in explaining order flow. Further, these effects will exist not just for non-S&P stocks, but also for S&P stocks.

³ Traders implementing index arbitrage strategies between the S&P cash and futures indexes will buy or sell baskets of stocks. Additional arbitrage possibilities have been created by the arrival of SPDRs.

We also study time-of-day patterns in commonality, since index funds are likely to be especially active at the end of the trading day.⁴ Finally, we compare the strength of common effects in 1986 and 1996, since the growth over this period in mutual fund assets, particularly assets in index funds, has been spectacular.⁵ If commonalities come from index fund trading, we should see a significant increase in commonality between 1986 and 1996, especially for S&P stocks.

The economic effects of order flow commonality. We also explore hypotheses concerning the effects of common order flow on prices. Aggregate order flow need not affect prices at all, as correlated pricing effects could be explained by (correlated) stock-specific order flow. This is the case, for instance, if each specialist sets prices using only order flow in his stock (e.g. Glosten and Milgrom, 1985). However, an incremental role for common order flow could plausibly arise if specialists learn from order flow in related stocks, as suggested in multi-asset microstructure models.⁶ In particular, it seems reasonable to assume that index fund trading is not motivated by private information.⁷ Then, if specialists can, by observing aggregate S&P order flow, tell when index funds are

⁴ Index funds have a clear preference for trading near the close of the market. First, they are able to minimize tracking error in their returns relative to their benchmark (e.g. the S&P 500), whose value is computed using closing prices. Also, by waiting for a better estimate of net flow and thereby being able to offset individual inflows and outflows occurring during the day, institutions can reduce the required amount of trading and their trading costs. Anecdotal evidence supports these arguments. In describing the daily activities of the Vanguard 500 Portfolio, the *Wall Street Journal* (January 28, 1997) reports that the fund “has been buying some stocks all day” but that “the bulk of its buying will occur in the next 45 minutes, right before the close of the New York Stock Exchange at 4:00 p.m.” In the same newspaper (March 3, 1997) the fund’s manager describes why one-third of his trades occur in the final hour: “[S]ince the investor is buying the fund at the closing price of the day, we want to pay a price as close as possible to that on close.”

⁵ Investment Company Institute data show that assets in stock funds totaled \$1.73 trillion in December 1996 compared to only \$117 billion in December 1985. Figures from Lipper show that assets in mutual funds linked to the S&P 500 grew from \$473 million to \$37 billion between December 1985 and December 1995, and that the number of such index funds rose from four to 43 over this period. According to more recent figures from the Investment Company Institute, assets in S&P 500 funds were \$155.7 billion in June 1999. The *Wall Street Journal* (mutual fund supplement, April 9, 2001) reports that 12% of stock mutual fund, and 25-30% of pension fund, assets are indexed. While this article doesn’t mention the percentages in S&P 500 funds, the discussion indicates that much of these assets are in S&P-linked funds.

⁶ Caballe and Krishnan (1994) assume that every trader and market maker can observe order flow in all assets, while Kumar and Seppi (1994) assume that this observability occurs with a lag. Easley, O’Hara and Srinivas (1998) assume that stock and options market makers are able to view trading activity and prices in the related market. Admati (1985) considers price setting in a multi-asset rational expectations model.

⁷ Subrahmanyam (1991) makes the point that private information is less likely to be an important motive for traders buying or selling portfolios of stocks. This point is strengthened for index funds whose trading is motivated solely by net fund flows aggregated over individual investor orders.

trading, they can use that information to revise the probability that a given trade in an S&P constituent is uninformed. The prediction is that a trade in an S&P stock made in the same direction as aggregate S&P order flow should have a smaller price effect than one in the opposite direction of aggregate order flow. Given the increasing importance of indexing through time, the effects should be stronger in 1996 relative to 1986, especially at the end of the day. It is also possible that market-wide order flow is informative to specialists, implying that broader market order flow could affect prices as well.

Hypothesis Testing. Our analysis of constituents of the S&P 500 and of the non-index universe measures the pervasiveness of common effects. By studying two widely spaced years, 1986 and 1996, we are able to study trends in common effects. Finally, our estimation strategy allows us to draw inferences about the relation between common effects in order flow and in returns.

Our approach, of identifying candidates for common factors and examining their explanatory power, differs from Hasbrouck and Seppi's use of factor analysis, which does not address the identity of the common factors. We calculate the average correlation in intraday (fifteen-minute) order flow for pairs of stocks in the S&P and non-S&P samples, and do the same for intraday returns. In order to identify the factors driving order flow commonality, we estimate a sequence of regression models for fifteen-minute order flow for S&P and non-S&P stocks, starting with firm-specific variables and moving to more general factors, reflecting industry, indexing and broad market order flow and (lagged) returns.⁸ After each regression, we examine residual commonality to gauge the power of each source to explain common effects.

To address the economic importance of order flow commonality we regress returns on own order flow and the common order flow factors. We proceed as with the regressions explaining order flow, progressively adding the industry, index and broader market order flow factors and measuring the changes in the regression R-squared and the residual correlation. Our final specification includes interactive terms, which help test

⁸ We construct the factors for the S&P 500, the non-S&P universe and for industry groupings.

the effects of common order flow on the price impact of order flow for an individual stock.

Finally, we use the method of principal components to statistically extract common factors for order flow and returns. By comparing these factors to our economically motivated factors, we are able to link our proposed factors and findings to previous findings based on statistically generated common factors.

3 Data and empirical methods

Our tests analyze stocks in the S&P 500 and a random sample of non-S&P firms for two years, 1986 and 1996. We choose two widely spaced years in order to capture possible effects associated with the growth in trading, especially institutional trading.⁹ The initial year is 1986 since the current trading hours on the NYSE (9:30 am to 4:00 pm) took effect in September, 1985. The later year is 1996 because it is the last year for which we were able to obtain the full list of index firms.

The list of firms in the S&P 500 is obtained from biannual issues of the S&P *Stock Price Record*. In order to obtain a sample of non-S&P stocks, we exclude ADRs, closed-end funds, REITs and sort the remaining NYSE-listed common stocks into quintiles on the basis of annual trading volume for each year. We then select the first 75 firms per quintile in alphabetical order. This procedure is designed to produce a random sample of firms across industries, while obtaining broad representation in terms of trading activity. Before selecting firms but after defining the volume quintiles, we drop firms that in either year (a) have an average price below \$10; and (b) have fewer than 100 trades in total. The purpose of the first condition is to exclude firms for which microstructure problems are likely to be severe, e.g. volatile returns. That of the second is to rule out firms that are very infrequently traded (have less than one trade every two days).

⁹ While figures on institutional trading are difficult to locate, institutional ownership has grown enormously. For instance, the NYSE Factbook (2000) states that institutional ownership grew from 41.7% in 1990 to 50.8% in 1995. This increased ownership is likely to be associated with more intensive institutional trading.

Transaction and quote data come from the Institute for the Study of Securities Markets (ISSM) tapes for 1986 and from the Trade and Quote (TAQ) CDs for 1996. Following Hasbrouck and Seppi (2001), we measure variables over fifteen-minute intervals. The presumption in our tests and those in Hasbrouck and Seppi, and indeed underlying theoretical models of market microstructure, is that causation runs from trading activity to prices.¹⁰ Accordingly, we would like to be able to attribute changes in prices to patterns in order flow. However, as the measurement interval is lengthened there is a heightened risk of causality being confounded by feedback from prices to trading activity. The use of fifteen-minute intervals represents a balance between minimizing such endogeneities and maximizing the number of intervals with non-missing observations. To examine time-of-day effects, we separately analyze the four fifteen-minute intervals corresponding to the last hour and the 22 intervals corresponding to the rest of the day. If correlated trading effects are driven by institutional transactions and institutions prefer to trade near the end of the day, these effects should be striking at the end of the day.

To calculate order flow, we first sign each trade by comparing the transaction price with the midpoint of the prevailing quote, as long as the quote has been in effect for at least five seconds when the trade occurs (see Lee and Ready, 1991). Buys, given a positive sign, are transactions occurring at prices above the midpoint, while sells, given a negative sign, are transactions occurring at prices below the midpoint. We exclude trades occurring at the quote midpoint. Order flow can be defined in terms of the number of transactions (the difference between the number of buys and the number of sells) or in terms of trading volume (which also incorporates the number of shares bought and sold). Prior research, e.g. Hasbrouck (1991), finds that order flow based on the number of transactions is better behaved. Hence, the results that we report use this measure of order flow. The number of transactions is cumulated for each fifteen-minute interval in the day.

¹⁰ This is exemplified in the common result that the specialist learns from order flow, and adjusts prices to reflect, in an unbiased manner, the information in order flow.

Intraday returns are calculated using the average of the bid and ask prices, so as to avoid bid-ask bounce. Returns are based on the most recently available midpoint, starting with the first quote of each day, and will automatically be zero if a quote has not been posted in any interval. We address the possible effects of stale quotes on our results in the next section.

In order to construct the factors—aggregate S&P 500, non-S&P and industry order flow and returns—we compute order flow and returns for *all* NYSE-listed stocks on ISSM and TAQ for a particular fifteen-minute interval, and then aggregate these variables for the groupings of interest.¹¹ After excluding firms that fail to surpass the price and trade exclusionary hurdles as well as non-NYSE firms, we are left with 1196 firms in 1986 and 1549 firms in 1996. Of these, 383 firms belong to S&P 500 sample and 813 to the non-S&P sample in 1986; the corresponding sample sizes for 1996 are 454 and 1095. In only one industry (two industries) in 1996 (1986) is the number of firms less than 10. Otherwise, the number of firms in the industries ranges between 37 and 179 in 1986 and between 33 and 269 in 1996.

Aggregate order flow for each fifteen-minute interval is the sum of the order flows for the constituent stocks. The aggregate return is an equally weighted cross-sectional average for the interval, using only stocks with actual quotes in the interval (to prevent stale prices from affecting aggregate returns).

4 Results

4.1 *Trading activity: descriptive statistics*

Table 1 provides descriptive statistics for the S&P 500 and non-S&P samples for 1986 and 1996.¹² The figures in this table are fifteen-minute averages, and are calculated separately for the first 5½ hours and the last hour of the trading day. Starting with

¹¹ We group stocks into 16 industries. The industry classifications are available upon request.

¹² Since we do not include NASDAQ stocks, and are missing data for a few stocks, the figures do not represent the entire S&P 500. We will abuse the term, for the sake of brevity. Also, the statistics for the non-S&P sample in table 1 are calculated using *all* non-S&P stocks, not just the 375 stocks used in our tests.

trading activity for 9:30 am to 3:00 pm, total trading volume for S&P stocks has increased from 26,000 round lots to almost 70,000 round lots per fifteen-minute interval over this ten-year period. The increase in non-S&P trading volume is also substantial though not as large. Order flow (measured using net volume or net number of trades) is, for the most part, positive for both groups, and has grown substantially between 1986 and 1996. Per interval trading volume and order flow are larger during the final hour than during the rest of the day, in each year and for both samples. The intraday patterns in trading activity are consistent with the tendency of institutions to place trades near the end of the day.¹³

The descriptive statistics show that trading activity has increased overall, especially at the end of the day, and with slightly larger increases occurring for stocks in the S&P 500. They suggest that we are most likely to find evidence of order flow commonality and of its effects at the end of the day.

4.2 *The magnitude of common effects: correlations*

Before addressing the hypothesized sources of common effects, it is important to examine their strength and pervasiveness. To document these common effects, we calculate correlations in fifteen-minute order flow (net number of trades) and returns for every pair of stocks in the S&P and non-S&P samples and for 1986 and 1996. The correlations are calculated using the time-series of observations for each fifteen-minute interval (e.g. 10:30-10:45 a.m.) and then averaged across the two broad intervals corresponding to the first 5½ hours and the last hour of the trading day.¹⁴ Order flow is

¹³ We should comment on the fact that order flow is positive in both years. As a percentage of total volume or number of trades, this order imbalance is small, less than 5% in each case. We also compare our numbers with descriptive statistics in table 2 of Hasbrouck and Seppi. Their mean net volume is 16 round lots per firm per fifteen-minute period in 1994. Averaging net volume for the longer interval, 9:30 am to 3:00 pm, across the number of S&P 500 firms in 1996 (454), we get approximately five round lots per firm. Hasbrouck and Seppi's figures probably are larger because they look at the thirty Dow Jones stocks.

¹⁴ A detailed analysis of the correlations by fifteen-minute interval reveals relative homogeneity within the last hour and within the rest of the day, suggesting that the coarser partition is reasonable. In particular, the correlations during the opening hour are lower than those corresponding to the last hour, and are not different from those corresponding to the rest of the day.

cumulated within two trade size categories, 100-9,900 shares and more than 10,000 shares (large order flow).

Several general statements can be made about the order flow and return correlations in table 2. The pairwise correlations for both are appreciably larger for S&P stocks than for non-S&P stocks, and are mainly in excess of 10% for S&P stocks. Given the high-frequency nature of the data, these correlations are substantial and would be expected to increase as the interval was lengthened to a day or longer. For S&P stocks, the correlations increase at the end of the day and from 1986 to 1996. Finally, the correlation for large order flow is insignificant in both years for S&P stocks (this also holds for non-S&P stocks). This is not surprising since block trades tend to be infrequent and often executed in time-consuming ways. Therefore, in subsequent tests, we drop all block trade order flow and focus on order flow occurring for transactions between 100 and 9,900 shares.

These results provide initial support for the indexing hypothesis. The facts that the order flow correlations are economically large for S&P stocks, and are higher at the end of the day and in 1996 are consistent with strong common order flow being induced by index funds trading especially intensively at the end of the day. Note the positive correlations for S&P stocks during the rest of the day. This is consistent with the actions of index arbitrageurs (which need not be confined to the end of the day) or with funds spreading out their trades to avoid placing all of their trades at the end of the day. Given the growth in indexing through time, funds are more likely to spread their trades throughout the day in 1996, accounting for the larger increase in the order flow correlation during the 9:30 a.m.-3:00 p.m. interval. However, the correlation remains below that at the very end of the day. The return correlations, which mirror the pattern of the order flow correlations, indicate the material price impact of order flow. The elevated correlation at the end of the day suggests that institutional transactions have especially pronounced effects on prices at this time.

Table 2 also shows that the order flow and return correlations in the non-S&P sample are less than 2%. The small correlations imply that common effects are economically unimportant for non-index stocks. Since previous evidence in Hasbrouck and Seppi

(2001) is based on only the widely tracked Dow stocks, this finding is significant in that it indicates that commonality is basically an index-driven phenomenon. We will, however, evaluate the ability of broader market factors to explain common effects in the non-S&P sample.

We check that the results in table 2 are not mechanically produced by differences in trading activity across the two samples, or across the two years. If non-S&P stocks trade infrequently, many fifteen-minute intervals will have zero order flow and zero returns. This could lead to lower correlations for non-S&P stocks than for S&P stocks. Similarly, the correlations could be lower for 1986 than 1996 simply because trading activity has increased between the two years. As a quick test of this possibility, we ignore all intervals with no trades or quotes and recompute the correlations using only the remaining ‘active’ intervals. If non-trading or stale prices are driving the observed patterns, we should see a dramatic increase in the pairwise correlations for the non-S&P sample relative to the S&P sample and for 1986 relative to 1996. The results (not tabulated) continue to support all of the inferences from table 2.

In addition, we calculate hourly order flow correlations. If asynchronous trading is a problem, then we expect to see the correlations increasing as the measurement interval is lengthened. The non S&P order flow correlations increase slightly (to approximately 3%) for all intervals. The S&P order flow correlations increase to 16% at the end of the day and 8% for the rest of the day in 1986, but increase less substantially in 1996, consistent with heavy trading in index stocks by 1996. However, our earlier conclusions regarding the differences across samples and through time are unaltered.

To summarize, we uncover significant common effects in order flow and returns for stocks in the S&P 500. Common effects are substantially weaker, and economically inconsequential, for a broader sample of randomly selected stocks. Together, the results suggest that index-related transactions are important drivers of common effects. They also prompt two follow-up questions: first, can we directly identify some of the forces behind the common effects in order flow; and second, do the common effects in order flow cause those in returns? We address these questions in the next two sections.

4.3 Explaining order flow commonality

Our next task is to explain the common effects in trading activity documented in table 2. We proceed by specifying the following regression between order flow for each stock and aggregate factors and own-stock control variables:

$$\begin{aligned} OF_{it} = & a_0 + a_1 OF_{i,t-1} + a_2 RET_{i,t-1} + b_1 OF_{Ind,t} + b_2 OF_{Ind,t-1} + b_3 RET_{Ind,t-1} + c_1 OF_{SP500,t} \\ & + c_2 OF_{SP500,t-1} + c_3 RET_{SP500,t-1} + d_1 OF_{Other,t} + d_2 OF_{Other,t-1} + d_3 RET_{Other,t-1} + e_{it} \quad (1). \end{aligned}$$

Here, i represents an individual stock, OF stands for fifteen-minute order flow (net number of trades) consisting of trades between 100 and 9900 shares, and t represents a particular fifteen-minute period; $OF_{SP500,t}$, $OF_{Other,t}$ and $OF_{Ind,t}$ are order flow for all S&P 500 stocks, non-S&P stocks, and the stock's industry in period t ; $RET_{SP500,t-1}$, $RET_{Other,t-1}$ and $RET_{Ind,t-1}$ are S&P, non-S&P and industry returns in the previous fifteen-minute period; $OF_{SP500,t-1}$, $OF_{Other,t-1}$ and $OF_{Ind,t-1}$ are the previous period's order flow; and $OF_{i,t-1}$ and $RET_{i,t-1}$ are the previous period's order flow and return for stock i .¹⁵

In order to allow comparisons across stocks and through time, we standardize each variable (stock-level and aggregate-level) by its standard deviation for the fifteen-minute period (calculated over the entire year). The regression is estimated separately for each stock and in each year using the time-series of standardized variables corresponding to the last hour of the day (i.e. four fifteen-minute periods for each day) and the remaining 5½ hours (i.e. twenty-two fifteen-minute periods per day).¹⁶ We then average the regression coefficients separately for stocks in the S&P 500 index and those not in the index.

We report the final specification (1), but estimate the model sequentially, starting with the own stock variables and then adding the industry, indexing and broader market return and order flow factors. The purpose is to examine the incremental contribution of the three sets of common factors by examining changes in R-squared and residual order flow correlation. However, since the incremental explanatory power attributable to a set

¹⁵ When using stock i 's order flow as the dependent variable, we remove stock i 's order flow from all of the appropriate aggregate order flow regressors.

¹⁶ Thus, for a stock in each year, there are approximately 250*4 observations at the end of the day and 250*22 observations for the rest of the day.

of regressors depends on the order in which they are added to the model, we estimate the model with different permutations of the common variables. For instance, in one permutation, we add the industry variables first, followed by the S&P variables and then the broader market variables. In a second permutation, we add the index variables first, followed by the industry variables and the broad market variables. The final specification is presented in table 3 and the performance of the sequence of models is summarized table 4.

The first panel of table 3 presents the median regression coefficients for the S&P 500 sample for the last hour of the day and the rest of the day, and the second panel presents the matching coefficients for non-S&P stocks. In the discussion of table 3 (and table 5), we will often make comparisons between coefficients. We have conducted formal tests of differences, but do not wish to complicate the presentation by tabulating the results of so many tests. Any difference we highlight is statistically significant and, in our opinion, economically significant.

S&P stocks. Our purpose is to test hypotheses about several non-mutually exclusive sources of commonality. The indexing hypothesis is generally supported by the results. The coefficient on contemporaneous aggregate S&P 500 order flow is always significantly positive and several times as large as the coefficient on the first lag of S&P order flow (which is negative). This means that there are more trades in an index constituent stock when other S&P 500 stocks are being traded, and these trades are of the same sign (buys or sells). The increase in the strength of this co-movement at the end of the day, when index funds are most active, also points to indexing as the underlying cause. The fact that the end-of-day coefficient is constant from 1986 to 1996 may appear to be at odds with indexing effects. However, the coefficient does not have to increase. Increased basket trading by index funds coupled with a constant per trade effect will still produce the observed stronger correlation in order flow for index constituents.

The performance of the index provides another source of commonality, with a lag. Two factors that motivate the inclusion of lagged index returns are index arbitrage strategies, determined in part by cash index returns, and feedback effects from returns to

trades. While index arbitrage effects should lead to a negative coefficient on index returns, positive feedback trading effects should lead to a positive coefficient, *ceteris paribus*.¹⁷ The positive coefficient on S&P returns from the previous fifteen-minute period is consistent with feedback effects. These effects are similar during the last hour and the rest of the day, which is reasonable since feedback trades need not exhibit time-of-day sensitivity.

The next hypothesized source of common effects is broad market shifts in allocations, preferences or information. Such commonality would present itself in the form of significant coefficients on aggregate non-S&P order flow and returns. The coefficients on the broader market return factor are almost always insignificantly different from zero. In 1986, the coefficient on aggregate non-S&P order flow is positive but small and that on the first lag is negative and of similar magnitude, and these coefficients are either insignificantly different from zero or negative in 1996. Moreover, the size of these coefficients is one-twentieth the size of the aggregate S&P coefficients. Thus, broad market factors are not important drivers of correlated order flow for index stocks.

Finally, the specification sheds light on the industry effects hypothesis. By its nature, this source of commonality is more localized, since it will only explain common effects within an industry. Nonetheless, there is evidence of commonality induced by industry order flow, although its strength is no more than one-tenth that of the indexing effect. Also, the positive coefficient on lagged industry returns suggests feedback trading. The increase over time in the coefficients shows that industry effects have become a more important source of order flow commonality in recent years.

Table 4 presents the trend in the residual correlation when we sequentially add common variables. An average residual correlation close to zero means that much of the commonality has been eliminated and is an indication of model adequacy. Note first that adding firm-specific variables (own order flow and lagged returns) does nothing to remove comovement. The residual correlations after firm-specific order flow and returns

¹⁷ For instance, positive cash index returns result in the cash index being overvalued relative to the futures index, holding futures returns constant. This should prompt selling in the stock market.

are added actually increase slightly in comparison to the raw correlations in table 2. This is not necessarily surprising since firm-specific variables should simply serve to dilute common effects. Hence, eliminating the effects of firm-specific variables heightens common effects. However, adding industry variables leads to a substantial reduction in the correlation, e.g. from in excess of 12% to under 2% in 1996. Following the addition of S&P order flow and returns, the correlation declines further to 0.1%, and is no longer significant. No additional reduction occurs when non-S&P variables are included. Table 4 also shows that when we add the S&P variables first, the residual correlation immediately drops to 0.1%, and the industry and non-S&P variables offer no further reduction. Thus, the major contributors to the reduction in order flow commonality for index stocks are S&P, and secondarily industry, order flow and returns.

Non-S&P stocks. The regression taking order flow in individual non-S&P stocks as the dependent variable provides evidence on the broad market effects hypothesis of marketwide, non-index-related commonality. While the correlations for non-S&P stocks in table 2 are low, it is still important to examine the influences for these stocks and to see if our hypothesized factors are able to explain common effects in this sample as well.

The coefficient on aggregate non-S&P order flow is reliably positive, though the coefficient is 30% as large as the comparable aggregate S&P coefficient in the regressions for S&P stocks. Thus, while broad market effects have a smaller economic impact, non-S&P stocks clearly have their own common influences, perhaps due to aggregate flows to stock funds in general or to broad allocation shifts on the part of individuals and institutions. The effect of aggregate non-S&P order flow on order flow for a non-S&P stock increases slightly at the end of the day, possibly due to institutional trading interest in all stocks. However, this effect does not increase through time. The effect of lagged broader market returns is positive but not significant in 1996. Moreover, the effects are weaker than those of index returns on index constituents.

Industry effects are present for non-index stocks as well. The effect of industry order flow on stock order flow is generally positive and significant, and that of returns is mixed in 1986, but positive in 1996. The industry order flow coefficients are comparable to those in the S&P regressions.

Finally, note that order flow for non-S&P stocks is negatively affected by aggregate S&P order flow and largely unaffected by lagged S&P returns. While the negative coefficient supports the intuition that indexing order flow spills over negatively to order flow for non-index stocks, the effect is only one-third as large as that of broader market order flow.

As with the S&P sample, we estimate the models sequentially in order to gauge the role of the common factors in accounting for order flow commonality and we present the results in panel B of table 4. Following the addition of the firm-specific variables, the residual correlation does not decline in relation to the raw correlation. Adding the industry factors next, we find a drastic reduction in commonality. Unlike the S&P sample, adding the index variables does not lead to further reductions in common effects. In fact, it requires the addition of the broader market factors to drive residual commonality to levels close to zero. Reversing the order of addition, we find that when the broader effects are added first, the residual correlation drops to a level very close to zero. Thus, for non-index stocks, broader market and industry factors (depending on the order of addition) play the decisive role in explaining correlated order flow. This conclusion accords well with our intuition that common order flow effects in the broader sample of stocks, while weaker than those for index constituents, are associated with market-wide and industry factors. We note that the residual correlations of S&P stocks when indexing effects are removed first are on par with the raw order flow correlations for non-S&P stocks (from table 2). Once again, this is consistent with most of the order flow commonality in S&P stocks being due to indexing effects, which are largely absent in order flow for non-S&P stocks.

As a final comment, note that table 3 also shows that the explanatory power of the regressions is greater for S&P stocks. The specifications show an ability to explain 20% (13%) of the order flow variation for individual index stocks during the end (rest) of the day in each year, but only approximately 5% of the order flow variation for individual non-S&P stocks. The higher explanatory power for index constituents reflects both the strong effects of indexing and also, possibly, the fact that indexing factors are easier to capture than are broader market factors. Overall, our results demonstrate that the proxies

for our hypothesized common factors—index, industry and broad market order flow and returns—completely explain commonality in order flow. An additional benefit of the zero residual correlations is that the statistical tests on the coefficients, which assume independence, are correctly specified.¹⁸

4.4 *The economic effects of order flow commonality*

Having established the existence of common factors in order flow, we next examine their effects on prices. The main goal of this analysis is to see if common order flow has incremental effects. A secondary goal is to assess whether the results are consistent with the predictions of certain economic models.

We carry out the analysis in the context of a regression specification that relates returns to a stock's own order flow:

$$R_{it} = a_{it} + b_{it}OF_{it} + e_{it} \quad (2),$$

where

$$a_{it} = a_{i0} + a_{i1}OF_{SP500,t} + a_{i2}OF_{Other,t} + a_{i3}OF_{Ind,t}; \text{ and}$$

$$b_{it} = b_{i0} + b_{i1}OF_{SP500,t} + b_{i2}OF_{Other,t} + b_{i3}OF_{Ind,t}$$

are time-varying regression coefficients for stock i , R_{it} is the return for stock i in fifteen-minute period t , OF_{it} is period t order flow for stock i , and $OF_{SP500,t}$, $OF_{Other,t}$ and $OF_{Ind,t}$ are aggregate order flow for S&P 500 stocks, non-S&P stocks and the stock's industry in period t . Substituting the definitions of a_{it} and b_{it} into (2) leads to the following relation:

$$R_{it} = a_{i0} + a_{i1}OF_{SP500,t} + a_{i2}OF_{Other,t} + a_{i3}OF_{Ind,t} + b_{i0}OF_{it} + b_{i1}OF_{it}OF_{SP500,t} + b_{i2}OF_{it}OF_{Other,t} + b_{i3}OF_{it}OF_{Ind,t} + e_{it} \quad (3),$$

¹⁸ Summarizing the effects of own stock variables, the coefficient on lagged stock returns is negative and significant. This is consistent with prior evidence (e.g. Hasbrouck, 1991), and suggests profit-taking or inventory control (or both) at high frequencies. The positive coefficient on lagged order flow suggests that buys tend to be followed by buys, and is typical of high frequency data (see, for instance, Hasbrouck, 1991).

a specification in which returns for a stock are related to aggregate S&P, non-S&P and industry order flow, order flow for the stock, and stock order flow interacted with aggregate S&P, non-S&P and industry order flow. The regressions are estimated separately for the last hour of the day and the rest of the day, in order to examine whether the effects are different at the end of the day, when institutions are likely to be more active.¹⁹

The time-varying nature of (2) and (3) allows us to explore whether the specialist learns from order flow in assets besides his own, as suggested in multi-asset microstructure models. Specifically, in the extreme, if a specialist conditions prices on order flow in his stock, paying no attention to order flow in other stocks, only b_{i0} (and possibly a_{i0}) will be non-zero. On the other hand, if the specialist adjusts prices on the basis of order flow in other stocks, e.g. index components, one or more of the other coefficients will be non-zero, depending on which order flow is informative to the specialist. For instance, if aggregate S&P order flow directly causes revisions in the value of stock i , a_{i1} will be non-zero. Similarly, b_{i1} will be non-zero if there is an interaction effect between order flow for the entire S&P 500 and for stock i .

Table 5 presents the results for specification (3). The results are summarized for S&P and non-S&P stocks, for the last hour and the rest of the day, and for 1986 and 1996. Positive (negative) order flow in a stock leads to positive (negative) returns (column 2). Columns 4 through 12 demonstrate the effects of order flow commonalities. In cases where the direct and interaction effects are of opposite signs, the interaction effect dominates for most reasonable levels of order flow, and our discussion presumes a dominant interaction effect.

S&P stocks. In 1986, the negative coefficient on the interaction of stock order flow and S&P order flow implies that the price effect of own order flow is smaller when order flow in the index is of the same sign, presumably because such trades are more likely to

¹⁹ In estimating (3), we do not remove own stock order flow from aggregate order flow. This seems reasonable since we include own order flow as a separate regressor. We also add the first lag of each order flow variable to see if there are lags in their effects on prices. These terms are suppressed in (3) in order to save space.

be part of a basket trade and hence uninformed. Conversely, if order flow in a stock is of the opposite sign to that in the index, the price effect is larger, due to the increased likelihood that it represents informed trading. By 1996, however, the interactive effect is no longer significant. A possible explanation for the 1996 result is that the sheer weight of index fund trading leads to inventory-induced price adjustments in the direction of trading.²⁰

By contrast, the effect of non-S&P order flow on the price effect of stock order flow is positive throughout the day in 1986, and at the end of the day in 1996. The effect of industry order flow is likewise positive in both years, consistent with industry-level information effects.

Table 6 reports the residual return correlation when we estimate the model sequentially, starting with own order flow and then adding industry, S&P and broad market order flow (and the appropriate interaction effects). If the return correlation is explained by (correlated) stock order flow, the residual return correlation will be close to zero after we add stock order flow. Similarly, if a particular common order flow factor fully explains the correlation in returns, the residual correlation will be zero after we add this factor.

As shown in panel A of table 6, the addition of stock order flow leads to a reduction of between 50% and 60% in the return correlation, for instance, from 14% to 6% at the end of the day in 1996. The addition of industry order flow results in another substantial drop in the correlation, to approximately 2% in each year. Adding S&P and non-S&P order flow does little to lower these correlations any further. When we add index order flow first followed by industry order flow, the return correlation displays a matching decline at each step. The similar influences of industry and index order flow shows that both are important in explaining return comovement. However, own order flow seems to be the principal driver of the return correlation across stocks.

²⁰ Note that the direct effect of basket trading is to move prices in the direction of order flow, as seen in the positive coefficient on aggregate S&P order flow and/or its first lag. The remaining lagged order flow coefficients in both panels are dominated by the contemporaneous order flow effect.

Non-S&P stocks. The regression results are shown in panel B of table 5. The negative coefficient on the interaction of order flow and non-S&P order flow implies the presence of a basket discount for non-S&P stocks in both years except at the end of the trading day. The absence of a discount at the end of the day is consistent with institutional trading containing information about prices or imposing inventory risk on specialists. Also, aggregate S&P order flow has noticeable price effects in the broader market. The net effect of stock order flow and S&P order flow of the same signs is positive in 1986 and during 9:30 a.m. through 3:00 p.m. in 1996, but is negative at the end of the day in 1996. The effect of industry order flow on prices is positive throughout the day and in each year.

Examining the stage-wise residual correlation in panel B of table 6, we see that own stock order flow brings the return correlation down to less than 0.5% in 1986, though it is less successful in explaining return comovement in 1996. When industry order flow or broader market order flow is added next, the return correlation declines by a further 50% in each year, and the remaining order flow variables (including S&P order flow) do not lower the residual correlation any further. Thus, own order flow followed by industry and broader market order flow explains return comovement for non-index stocks.

Summary. The return results do not consistently line up with any one set of predictions but the most important aspect of these results is that common order flow matters in price determination. The evidence in table 5 shows that the strong order flow commonalities in S&P stocks and the weaker commonalities in the broader market have economic importance: they impact prices of individual stocks, even after controlling for the stock's own order flow. Consistent with the hypothesis that specialists use order flow information from other stocks when reacting to trades in their stock, aggregate order flow significantly affects stock returns and, mirroring the growing importance of common trading, this effect has generally grown stronger over time. With the exception of S&P stocks in 1986, the average regression R-squared is in excess of 20% in both samples of stocks. Finally, the fact that the pairwise residual correlations are no more than one-fifth of the raw return correlations presented in table 2 indicates that much of the return comovement can be attributed to order flow.

4.5 *Principal components*

In this section we estimate principal components in order flow and returns for both samples of stocks. Most importantly, we can then compare the extracted principal components with our economic factors to see if there is any correspondence. In addition, we extend the analysis of Hasbrouck and Seppi (2001) to wider samples.²¹

We have a pooled time series, cross sectional dataset with stock level order flow for 26 intervals per day for two separate years. With principal components, we estimate a set of variables (or factors), smaller than the number of stocks, that collectively maximize the overall explained variation in the series of stock order flow (this is equivalent to least squares, though pooled across stocks and time intervals). Specifically, principal components are mutually orthogonal linear combinations of the stock order flows for each interval, with constant weights, that accomplish this goal. We estimate the principal components for both order flow and returns for each group of stocks, e.g. S&P constituents, and for each year. We estimate the principal components from the correlation matrix of the stock order flows or returns. As a result, the total variation in each sample is the number of stocks, e.g. 454 in the S&P sample in 1996. Each eigenvector of the correlation matrix represents an extracted principal component, and the corresponding eigenvalue (scaled by the number of stocks) provides the fraction of total variation explained by that principal component.²²

Table 7 presents the first five principal components for the two samples, the two years and the two broad time periods in the day. Panel A provides the order flow results, and shows that the first principal component stands out in each year for S&P stocks.²³ In

²¹ Hasbrouck and Seppi also estimate canonical correlations to study covariation in linear combinations of returns and order flow. We do not estimate canonical correlations, because our goal here is to assess the extent to which our economic factors are statistically optimal.

²² Jobson (1992) provides a good textbook discussion of principal components.

²³ Assuming a sample of n observations drawn from a multivariate normal population, the standard error of the i th principal component, λ_i , is $\sqrt{2\lambda_i^2/n}$ (e.g. see Johnson and Wichern, 1992, p. 381). Since n is approximately 5500 (250 days multiplied by 22 intervals per day), the standard error of the first principal component in order flow for the S&P sample in 1986 (during 9:30 a.m.-3:00 p.m.) is $\sqrt{2 \times 34^2 / 5500}$, or 0.648.

Thus, the first principal component is highly significant. When we calculate the standard errors of the remaining principal components, we find that many are statistically significantly different from 1.0.

1986, it explains 9% of total variation during 9:30 a.m.-3:00 p.m. and 16% during the last hour of the day. In 1996, the explained fractions rise to 12% and 18%. The remaining principal components are appreciably smaller in each year, explaining in the region of 1% of total order flow variation. In each year, the first principal component explains a larger fraction of order flow variation at the end of the day. In the non-S&P sample, the difference between the first and subsequent principal components is not pronounced. The first component explains less than 3% of order flow variation in each year, and there is hardly any increase at the end of the day. Likewise, there is no increase in the importance of the first principal component through time.

The principal components in returns are presented in panel B of table 7. For S&P stocks, the first principal component is large, explaining 5% (9%) of return variation during the first 5½ hours (last hour) of the day in 1986 and 13% and 16% in 1996. The remaining components account for less than 1% of return variation. In the non-S&P sample, the first principal component never explains more than 1.5% of return variation, and yet dwarfs the remaining components. Once again, the patterns in the principal components in returns resemble those in order flow, across samples, years and the time of day.

Next, we examine the correlation between the principal components and the economic factors. Our motivation for doing so is to attempt to identify the statistically extracted principal components. Table 8 shows the correlation between the first five principal components and the indexing and broader market factors. We do not consider the industry factors in this exercise since they have more localized effects.

Panel A of table 8 shows that for S&P stocks, the correlation between the first principal component in order flow and aggregate S&P order flow is 0.98 in 1986 and 0.99 in 1996. This component is also correlated (though less strongly so) with non-S&P order flow. Given the overwhelming contribution of the first principal component in panel A of table 7, its near-perfect correlation with index order flow reinforces the

Since our focus is the economic significance of the principal components, we concentrate on the fraction of explained variation.

economic importance of aggregate index order flow in driving common order flow for index constituents.

For non-S&P stocks, the correlation between the first principal component in order flow and aggregate non-S&P order flow is 0.92 in 1986 and 0.95 in 1996. The third principal component has a correlation of 0.26 with aggregate index order flow in 1986 but this reverses sign in 1996. The remaining correlations are much smaller. Thus, the dominant principal component can be identified as aggregate broad market order flow.

Panel B of table 8 provides the correlation results for the first five principal components extracted from returns for the two samples of stocks. For S&P stocks, the first principal component is highly correlated with aggregate S&P order flow (0.86 in 1986 and 0.94 in 1996), while the second principal component is highly correlated with non-S&P order flow though only in 1986. The remaining correlations are not large. For non-S&P stocks, the correlation between the first principal component and non-S&P order flow is huge (0.85 in 1986 and 0.87 in 1996). Of the remaining correlations, only that between S&P order flow and the second principal component in 1986 is economically large. The correlations confirm that the statistical factors in returns are associated mainly with aggregate order flow for the sample under consideration.

Finally, we extract principal components from the residual order flow series and return series after controlling for the economic factors. The residuals are obtained from regressions (1) and (3), estimated in tables 3 and 5. If the economic factors truly explain common effects, the principal components should diminish substantially. Our untabulated results reveal this to be the case. For instance, the first principal component explains less than 1.5% of order flow variation in each year for index constituents and less than 0.75% for non-index stocks. Similar results hold for residual returns. Thus, our economic factors have succeeded in lowering statistically measured common effects in order flow and returns. (We do not estimate the correlation between the principal components extracted from the residual series and the economic factors because the residuals are orthogonal to the factors by construction.)

5 Discussion

This paper shows that substantial and growing commonalities exist in order flow and returns for stocks in the S&P 500 index. For these stocks, the striking order flow results which strengthen at the end of the day are consistent with the hypothesis that basket trading on the part of arbitrageurs and index funds is the dominant contributor to common effects in order flow. Additionally, periods when aggregate S&P order imbalances match those in the stock tend to be associated with improved prices.

However, common effects are not as substantial for stocks outside the index. This reinforces the idea that indexing drives common effects and indicates that commonalities, while important due to the market value of index constituents, are limited to a small part of the total population of stocks. The smaller commonalities that do exist in the non S&P sample are likely to be caused by concentrated institutional interest in particular industries or in firms meeting size and prudent investor screens. Further, because common order flow in an index is driven primarily by liquidity trades of indexers, it has a different effect on prices than does common order flow in stocks not part of an index.

The commonalities identified here have practical significance for market participants. Given the reality of correlated order flow and specialists' significant price reactions to common order flow, investors with discretion could time their transactions to take advantage of common order flow. For instance, an investor who wishes to buy shares in an index constituent might want to time this purchase to coincide with aggregate buys (since this leads to more favorable prices). This investor can use (public) information on the lagged S&P 500 return to determine when aggregate buys are more likely. Our results show that buys are likely after the index has risen in the previous period.

In addition, the relation between common order flow and prices of individual stocks is relevant to understanding the ability of investors to escape intraday downturns in groups of stocks. For instance, if the S&P is down in the face of selling pressure, are investors holding portfolios of non-index stocks affected? Assuming a zero order imbalance for non-index stocks, the generally positive (net) effect of aggregate S&P

order flow in the return regressions for non-S&P stocks shows that such investors will be affected adversely by selling in the index.

Besides having implications for the appropriate view of the specialist's information set, discussed earlier, our results also have implications for the debate on what drives return covariation. We find that order flow accounts for a huge fraction of return correlation. This leaves only a small fraction of return comovement to be explained by, for instance, pure public information (that moves prices without inducing order flow).

6 Conclusion

In this study we explore the causes and implications of common effects in order flow. Specifically, we examine the importance of indexing, industry and marketwide influences while controlling for own stock effects, and use membership in the S&P 500 to identify stocks that are subject to strong indexing effects. In addition, we analyze time-of-day and intertemporal effects since, relative to individuals, institutions account for more trading at the end of the day and in more recent years.

Our results show strong common effects in order flow for S&P 500 constituent stocks, and an increase in the extent of these effects at the end of the day and through time. Order flow for stocks in the S&P 500 is associated primarily with index order flow and lagged returns, and secondarily with industry order flow and returns. This evidence for S&P constituents is consistent with a leading role for institutional index trades in generating correlated order flow. Our results also show that statistically significant, though economically small, common effects exist in order flow for non-index stocks, driven by industry and broader market order flow and industry returns.

We explore the economic implications of order flow commonality for market participants. Returns are more strongly correlated across S&P 500 constituents than across non-S&P stocks. Furthermore, aggregate S&P, non-S&P and industry order flow has incremental explanatory power for returns. The commonality apparent in the raw order flow and return data largely disappears once the hypothesized factors are taken into account. Finally, principal components estimation reveals high correlations between the

dominant statistically extracted factors for order flow and returns in the two samples and aggregate index and broader market order flow.

Our analysis suggests that specialists monitor trading in related stocks and adjust the terms of trade accordingly. Additionally, our results related to the price effects of common order flow have implications for the optimal timing of trades. Finally, the striking high-frequency return correlations produced by order flow, particularly at the end of the day, are relevant to practitioners looking to diversify risk.

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Table 1: Summary statistics for S&P and Non-S&P stocks

Net and total volume and number of trades are presented for the two sample years, 1986 and 1996. Net-trades (net-volume) are calculated by classifying transactions as buys or sells (using the Lee-Ready, 1991, algorithm) and calculating the number of buys less sells (buy volume less sell volume) for each fifteen-minute interval. We use net trades as our order flow measure for the remaining tests in the study. Statistics are separately computed by S&P membership and by time of day. Volume numbers are round lots. The numbers presented are fifteen-minute averages; for example, there was an average total of 26225 round lots traded in all S&P stocks every fifteen minutes during the first part of the day in 1986.

Summary Statistics	1986		1996		% Change	
	9:30-3:00	3:00-4:00	9:30-3:00	3:00-4:00	9:30-3:00	3:00-4:00
S&P Net-Volume	938	1209	2408	4894		
Non-S&P Net-Volume	0	169	551	1661		
S&P Net-Trades	49	82	99	164		
Non-S&P Net-Trades	12	30	56	73		
S&P Total Volume	26225	28623	69575	87434	165%	205%
Non-S&P Total Volume	15208	16603	35366	44306	133%	167%
S&P Total Trades	1670	1784	2726	3852	63%	116%
Non-S&P Total Trades	1130	1151	1566	1985	39%	72%

Table 2: Order flow and return correlations

Median pairwise correlations for order flow (net number of transactions) and returns are tabulated. The numbers are presented for S&P and non-S&P stocks, for different times in the day, for different trade sizes (in the case of order flow), and finally for the two sample years, 1986 and 1996. The correlations are computed at the fifteen-minute interval level and grand averages are presented for the first 5½ hours of the day and the last hour of the day. For example, the average correlation between order flow in trades of 100 to 9,900 shares for any two S&P stocks in a fifteen-minute period was 0.061 in the first part of the day during 1986.

		S&P		Non S&P	
1986		Median		1986	Median
9:30-3:00	100-9900	0.061		100-9900	0.008
	10000+	0.000		10000+	0.000
	Returns	0.041		Returns	0.006
3:00-4:00	100-9900	0.119		100-9900	0.013
	10000+	0.000		10000+	0.000
	Returns	0.070		Returns	0.008
1996				1996	
9:30-3:00	100-9900	0.105		100-9900	0.011
	10000+	0.002		10000+	0.000
	Returns	0.117		Returns	0.017
3:00-4:00	100-9900	0.159		100-9900	0.013
	10000+	0.004		10000+	0.000
	Returns	0.142		Returns	0.017

Table 3: Order flow regressions

This table presents estimates of equation (1) in the text. The regressions are designed to uncover the sources of common influences in order flow. The specifications are estimated separately for S&P and non S&P stocks, for two broad intraday periods, and for the two sample years, 1986 and 1996. The specifications are estimated at the stock level for each year, separately using the time-series of fifteen-minute observations covering the hours 9:30 am-3:00 pm (22 observations per day) and the hour (3:00-4:00 pm) at the end of the day (4 observations per day). For example, the dependent variable in the first row is order flow for an individual S&P stock for a given fifteen-minute interval between 9:30 am and 3:00 pm in 1986. The cross-sectional medians of the regression coefficients and the adjusted R-square are presented along with their p-values for each sample, year and time of day. There are two main types of regressors: order flow and its first lag (OF and OF₋₁) and lagged returns (Ret₋₁). S&P, Non-S&P, Industry, and Own refer to the aggregate S&P, aggregate non-S&P, aggregate industry, and the stock's own data. Bold indicates significance at the 10% level or better.

Panel A		Intercept	Own	Own	Industry	Industry	Industry	S&P	S&P	S&P	Non S&P	Non S&P	Non S&P	Adj Rsq
1986 S&P			OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	
9:30-3:00	Median	-0.029	0.223	-0.125	0.018	0.003	0.004	0.253	-0.047	0.017	0.014	-0.015	-0.005	0.138
	p-val	0.000	0.000	0.000	0.000	0.155	0.001	0.000	0.000	0.000	0.000	0.000	0.024	0.000
3:00-4:00	Median	-0.033	0.216	-0.129	-0.003	0.006	0.006	0.376	-0.057	0.017	0.012	-0.011	-0.006	0.201
	p-val	0.000	0.000	0.000	0.182	0.048	0.246	0.000	0.000	0.000	0.000	0.011	0.261	0.000
1996 S&P														
9:30-3:00	Median	-0.020	0.167	-0.094	0.028	0.002	0.014	0.308	-0.056	0.027	0.000	-0.011	-0.003	0.146
	p-val	0.000	0.000	0.000	0.000	0.060	0.000	0.000	0.000	0.000	0.471	0.000	0.113	0.000
3:00-4:00	Median	-0.019	0.173	-0.093	0.035	-0.005	0.011	0.384	-0.064	0.025	-0.010	-0.007	-0.003	0.207
	p-val	0.002	0.000	0.000	0.000	0.204	0.000	0.000	0.000	0.000	0.002	0.014	0.468	0.000
Panel B		Intercept	Own	Own	Industry	Industry	Industry	S&P	S&P	S&P	Non S&P	Non S&P	Non S&P	Adj Rsq
1986 NonS&P			OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	OF	OF₋₁	Ret₋₁	
9:30-3:00	Median	-0.011	0.148	-0.070	0.010	0.001	0.003	0.000	-0.003	0.002	0.068	-0.002	0.004	0.039
	p-val	0.009	0.000	0.000	0.000	0.417	0.000	0.709	0.776	0.516	0.000	0.047	0.026	0.000
3:00-4:00	Median	-0.013	0.152	-0.084	-0.002	0.010	-0.005	0.002	0.004	-0.006	0.100	-0.020	0.009	0.055
	p-val	0.000	0.000	0.000	0.971	0.016	0.261	0.477	0.446	0.117	0.000	0.002	0.013	0.000
1996 NonS&P														
9:30-3:00	Median	-0.009	0.147	-0.080	0.017	0.003	0.009	-0.010	-0.009	0.006	0.083	-0.009	0.004	0.034
	p-val	0.289	0.000	0.000	0.000	0.252	0.000	0.004	0.000	0.005	0.000	0.000	0.018	0.000
3:00-4:00	Median	0.000	0.151	-0.083	0.029	0.003	0.009	-0.030	-0.006	0.002	0.099	-0.015	0.004	0.040
	p-val	0.827	0.000	0.000	0.000	0.745	0.009	0.000	0.221	0.569	0.000	0.000	0.221	0.000

Table 4: Marginal Impact of Hypothesized Factors on Common Effects in Order Flow

The table presents the average adjusted R-squared and the average correlation of the residuals from stepwise estimation of equation (1). Stock effects are added to the specification first, followed by the common factors in the indicated order. For example, the first set of numbers shows that accounting for only a stock's own effects (Own OF₋₁ and Own Ret₋₁), the residual order flows of the S&P stocks in 1986 have an average pairwise correlation of 7.7% during the main part of the trading day. This number can be compared to the average pairwise correlation of raw order flows for the same stocks presented in table 2 (6.1%). Proceeding down the column, once the industry factors (Ind OF₋₁, Ind OF, Ind Ret₋₁) are added to equation (1), the average correlation of the residual order flows drops to 2.1% and the fraction of an individual stock's order flow variation explained by the regressors increases from 5.6% to 10.6%. The second column shows the effect of adding S&P factors (S&P OF, S&P OF₋₁, S&P Ret₋₁) before the industry factors.

Panel A: S&P Stocks

1986 S&P						1996 S&P					
	Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq
Just Own OF			Just Own OF			Just Own OF			Just Own OF		
9:30-3:00	0.077	0.056	9:30-3:00	0.077	0.056	9:30-3:00	0.117	0.027	9:30-3:00	0.117	0.027
3:00-4:00	0.142	0.062	3:00-4:00	0.142	0.062	3:00-4:00	0.172	0.031	3:00-4:00	0.172	0.031
Add Industry			Add S&P			Add Industry			Add S&P		
9:30-3:00	0.021	0.106	9:30-3:00	0.001	0.137	9:30-3:00	0.018	0.109	9:30-3:00	0.001	0.144
3:00-4:00	0.026	0.157	3:00-4:00	0.001	0.201	3:00-4:00	0.019	0.167	3:00-4:00	0.001	0.205
Add S&P			Add Industry			Add S&P			Add Industry		
9:30-3:00	0.001	0.137	9:30-3:00	0.001	0.137	9:30-3:00	0.001	0.146	9:30-3:00	0.001	0.146
3:00-4:00	0.001	0.201	3:00-4:00	0.001	0.201	3:00-4:00	0.001	0.208	3:00-4:00	0.001	0.208
Add Rest of Mkt			Add Rest of Mkt			Add Rest of Mkt			Add Rest of Mkt		
9:30-3:00	0.001	0.138	9:30-3:00	0.001	0.138	9:30-3:00	0.001	0.146	9:30-3:00	0.001	0.146
3:00-4:00	0.001	0.201	3:00-4:00	0.001	0.201	3:00-4:00	0.001	0.207	3:00-4:00	0.001	0.207

Table 4, Panel B: Non S&P Stocks

1986 Non S&P						1996 Non S&P					
	Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq
Just Own OF			Just Own OF			Just Own OF			Just Own OF		
9:30-3:00	0.008	0.021	9:30-3:00	0.008	0.021	9:30-3:00	0.010	0.019	9:30-3:00	0.010	0.019
3:00-4:00	0.013	0.025	3:00-4:00	0.013	0.025	3:00-4:00	0.013	0.020	3:00-4:00	0.013	0.020
Add Industry			Add Rest of Mkt			Add Industry			Add Rest of Mkt		
9:30-3:00	0.002	0.033	9:30-3:00	0.000	0.039	9:30-3:00	0.002	0.031	9:30-3:00	-0.001	0.034
3:00-4:00	0.003	0.041	3:00-4:00	0.000	0.052	3:00-4:00	0.003	0.034	3:00-4:00	-0.001	0.038
Add S&P			Add Industry			Add S&P			Add Industry		
9:30-3:00	0.002	0.033	9:30-3:00	0.000	0.039	9:30-3:00	0.002	0.032	9:30-3:00	-0.001	0.034
3:00-4:00	0.004	0.042	3:00-4:00	0.000	0.054	3:00-4:00	0.003	0.035	3:00-4:00	-0.001	0.040
Add Rest of Mkt			Add S&P			Add Rest of Mkt			Add S&P		
9:30-3:00	0.000	0.039	9:30-3:00	0.000	0.039	9:30-3:00	-0.001	0.034	9:30-3:00	-0.001	0.034
3:00-4:00	0.000	0.055	3:00-4:00	0.000	0.055	3:00-4:00	-0.001	0.040	3:00-4:00	-0.001	0.040

Table 5: Return Regressions

Estimates of equation (3) are presented. OF stands for order flow (the net number of trades). The regressors are the stock's own order flow, industry order flow, aggregate order flow for all stocks in the S&P 500, aggregate order flow for non-S&P stocks, the first lag of these order flow variables (shown as OF₋₁), and the interactions between stock order flow and industry order flow, aggregate S&P order flow and non-S&P order flow (shown as Own*Ind OF, Own*S&P OF, Own*N S&P OF). The specification is estimated for each stock and for each year, separately using the time-series of fifteen-minute observations covering the hours 9:30 am-3:00 pm (22 observations per day) and the hour (3:00-4:00 pm) at the end of the day (4 observations per day). For example, the dependent variable in the first row is the return for an individual S&P stock in a given fifteen-minute interval between 9:30 am and 3:00 pm in 1986. The cross-sectional medians of the regression coefficients and the adjusted R-square are presented along with their p-values for each sample, for 1986 and 1996, and by time of day.

Panel A	Intercept	Own	Own	Industry	Industry	S&P	S&P	Non S&P	Non S&P	Own*S&P	Own*	Own*	Adj Rsq
1986 S&P		OF	OF ₋₁	OF	OF ₋₁	OF	OF ₋₁	OF	OF ₋₁	OF	N S&P OF	Ind OF	
9:30-3:00	-0.026	0.215	-0.022	0.008	0.000	0.056	0.024	0.103	-0.050	-0.024	0.004	-0.003	0.098
	0.000	0.000	0.000	0.000	0.430	0.000	0.000	0.000	0.000	0.000	0.033	0.434	0.000
3:00-4:00	-0.024	0.226	0.000	0.018	0.000	0.052	0.018	0.149	-0.088	-0.016	-0.005	0.000	0.138
	0.000	0.000	0.714	0.000	0.964	0.000	0.000	0.000	0.000	0.003	0.183	0.490	0.000
1996 S&P													
9:30-3:00	-0.048	0.378	-0.021	0.018	0.007	-0.015	0.062	0.138	-0.045	-0.002	-0.010	0.009	0.247
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.183	0.000	0.000	0.000
3:00-4:00	-0.052	0.349	-0.019	0.020	0.003	-0.019	0.056	0.169	-0.050	-0.001	0.003	0.000	0.253
	0.000	0.000	0.000	0.000	0.052	0.000	0.000	0.000	0.000	0.556	0.933	0.540	0.000
Panel B	Intercept	Own	Own	Industry	Industry	S&P	S&P	Non S&P	Non S&P	Own*S&P	Own*	Own*	Adj Rsq
1986 NonS&P		OF	OF ₋₁	OF	OF ₋₁	OF	OF ₋₁	OF	OF ₋₁	OF	N S&P OF	Ind OF	
9:30-3:00	-0.010	0.424	-0.009	0.002	0.000	0.019	0.029	0.013	0.002	0.002	-0.010	0.003	0.182
	0.000	0.000	0.000	0.037	0.372	0.000	0.000	0.000	0.352	0.962	0.000	0.258	0.000
3:00-4:00	0.000	0.426	0.000	0.006	0.001	0.015	0.030	0.015	-0.002	-0.003	-0.010	0.001	0.215
	0.921	0.000	0.549	0.066	0.896	0.000	0.000	0.000	0.331	0.479	0.084	0.758	0.000
1996 NonS&P													
9:30-3:00	-0.019	0.449	-0.005	0.008	0.005	-0.016	0.055	0.019	-0.007	0.013	-0.015	0.000	0.217
	0.000	0.000	0.012	0.000	0.004	0.000	0.000	0.000	0.000	0.000	0.000	0.752	0.000
3:00-4:00	-0.020	0.415	0.000	0.015	0.003	-0.023	0.052	0.019	-0.003	0.001	-0.004	0.001	0.195
	0.000	0.000	0.559	0.001	0.958	0.000	0.000	0.000	0.829	0.749	0.921	0.760	0.000

Table 6: Marginal Impact of Hypothesized Factors on Common Effects in Returns

This table is analogous to table 4, except that the stepwise estimation is performed for equation 3 from table 5. At each step we add the order flow and interactive terms in the order shown. For instance, in column 1, we add stock order flow first, followed next by industry order flow and industry order flow interacted with stock order flow, and then by S&P and non-S&P order flows and the associated interaction terms. The final specification corresponds to table 5.

Panel A: S&P Stocks

1986 S&P	Corr	Adj R-sq	Corr	Adj R-sq	1996 S&P	Corr	Adj R-sq	Corr	Adj R-sq
Just Own OF			Just Own OF		Just Own OF			Just Own OF	
9:30-3:00	0.026	0.068	0.026	0.068	9:30-3:00	0.060	0.201	0.060	0.201
3:00-4:00	0.035	0.089	0.035	0.089	3:00-4:00	0.062	0.192	0.062	0.192
Add Industry			Add S&P	Add S&P	Add Industry			Add S&P	
9:30-3:00	0.016	0.081	0.013	0.092	9:30-3:00	0.023	0.235	0.021	0.243
3:00-4:00	0.016	0.118	0.012	0.133	3:00-4:00	0.022	0.234	0.021	0.248
Add S&P			Add Industry		Add S&P			Add Industry	
9:30-3:00	0.013	0.093	0.013	0.093	9:30-3:00	0.020	0.245	0.020	0.245
3:00-4:00	0.012	0.134	0.012	0.134	3:00-4:00	0.020	0.251	0.020	0.251
Add Rest of Mkt			Add Rest of Mkt		Add Rest of Mkt			Add Rest of Mkt	
9:30-3:00	0.012	0.098	0.012	0.098	9:30-3:00	0.019	0.247	0.019	0.247
3:00-4:00	0.012	0.138	0.012	0.138	3:00-4:00	0.020	0.253	0.020	0.253

Table 6, Panel B: Non S&P Stocks

1986 Non S&P						1996 Non S&P					
	Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq		Corr	Adj R-sq
Just Own OF			Just Own OF			Just Own OF			Just Own OF		
9:30-3:00	0.003	0.176	9:30-3:00	0.003	0.176	9:30-3:00	0.011	0.206	9:30-3:00	0.011	0.206
3:00-4:00	0.004	0.195	3:00-4:00	0.004	0.195	3:00-4:00	0.010	0.178	3:00-4:00	0.010	0.178
Add Industry			Add Rest of Mkt			Add Industry			Add Rest of Mkt		
9:30-3:00	0.001	0.177	9:30-3:00	0.001	0.180	9:30-3:00	0.004	0.214	9:30-3:00	0.004	0.214
3:00-4:00	0.001	0.203	3:00-4:00	0.001	0.208	3:00-4:00	0.004	0.189	3:00-4:00	0.004	0.189
Add S&P			Add Industry			Add S&P			Add Industry		
9:30-3:00	0.001	0.181	9:30-3:00	0.001	0.179	9:30-3:00	0.004	0.214	9:30-3:00	0.003	0.215
3:00-4:00	0.001	0.206	3:00-4:00	0.001	0.215	3:00-4:00	0.004	0.192	3:00-4:00	0.003	0.194
Add Rest of Mkt			Add S&P			Add Rest of Mkt			Add S&P		
9:30-3:00	0.001	0.182	9:30-3:00	0.001	0.182	9:30-3:00	0.003	0.217	9:30-3:00	0.003	0.217
3:00-4:00	0.001	0.215	3:00-4:00	0.001	0.215	3:00-4:00	0.003	0.195	3:00-4:00	0.003	0.195

Table 7: Principal Components of Order Flow and Returns

The first five principal components of order flow and returns for S&P and Non S&P stocks are presented along with the fractions of total variation in order flow and returns that each explains. The analysis is performed separately for the two samples, the two intraday periods, and the two sample years.

	S&P					Non S&P				
	PC1	PC2	PC3	PC4	PC5	PC1	PC2	PC3	PC4	PC5
Panel A: Order Flow										
1986						1986				
9:30-3:00 Eigenvalue	34.23	2.80	2.66	2.10	2.00	6.64	2.64	2.20	1.98	1.89
% Explained	8.84%	0.72%	0.69%	0.54%	0.52%	1.87%	0.75%	0.62%	0.56%	0.53%
3:00-4:00 Eigenvalue	61.19	4.19	3.81	3.23	3.11	10.77	3.48	3.17	2.93	2.87
% Explained	15.81%	1.08%	0.98%	0.83%	0.80%	3.04%	0.98%	0.89%	0.83%	0.81%
1996						1996				
9:30-3:00 Eigenvalue	51.27	4.25	3.61	2.23	2.03	6.12	3.48	2.20	1.94	1.85
% Explained	12.12%	1.01%	0.85%	0.53%	0.48%	1.69%	0.96%	0.61%	0.54%	0.51%
3:00-4:00 Eigenvalue	74.91	5.57	4.49	3.05	2.96	7.75	4.72	3.09	2.76	2.62
% Explained	17.71%	1.32%	1.06%	0.72%	0.70%	2.15%	1.31%	0.86%	0.76%	0.73%
Panel B: Returns										
1986						1986				
9:30-3:00 Eigenvalue	20.88	3.18	1.83	1.69	1.65	5.25	2.25	1.74	1.67	1.62
% Explained	5.40%	0.82%	0.47%	0.44%	0.43%	1.48%	0.64%	0.49%	0.47%	0.46%
3:00-4:00 Eigenvalue	34.13	4.59	2.88	2.73	2.68	8.03	3.41	2.84	2.68	2.55
% Explained	8.82%	1.19%	0.75%	0.71%	0.69%	2.27%	0.96%	0.80%	0.76%	0.72%
1996						1996				
9:30-3:00 Eigenvalue	55.78	3.24	2.37	2.14	2.01	8.88	2.23	1.87	1.71	1.69
% Explained	13.19%	0.77%	0.56%	0.51%	0.48%	2.46%	0.62%	0.52%	0.47%	0.47%
3:00-4:00 Eigenvalue	66.20	4.15	2.92	2.84	2.65	9.46	2.98	2.77	2.66	2.61
% Explained	15.65%	0.98%	0.69%	0.67%	0.63%	2.62%	0.82%	0.77%	0.74%	0.72%

Table 8: Correlations of Principal Components and Hypothesized Sources of Common Effects

The first five principal components for order flow and returns are compared to the S&P and Non S&P (broad market) factors. Their correlations along with the p-values of those correlations are presented. Panel A presents the correlations between the factors and the principal components in order flow, and panel B the correlations between the factors and the principal components in returns. Bold indicates significance at the 10% level or better.

Panel A: Principal Components of Order Flow

	PC1	PC2	PC3	PC4	PC5		PC1	PC2	PC3	PC4	PC5
	S&P 1986						S&P 1996				
S&P OF	0.980	0.103	0.079	-0.001	-0.011	S&P OF	0.988	0.102	-0.011	-0.007	0.019
	<.0001	<.0001	<.0001	0.915	0.355		<.0001	<.0001	0.372	0.551	0.114
Non S&P OF	0.675	-0.010	0.093	0.069	0.036	Non S&P OF	0.692	0.086	0.023	0.021	0.028
	<.0001	0.423	<.0001	<.0001	0.004		<.0001	<.0001	0.062	0.083	0.025
S&P Ret ₁	0.385	-0.014	0.035	0.020	0.027	S&P Ret ₁	0.152	-0.134	0.099	0.040	0.001
	<.0001	0.269	0.007	0.126	0.039		<.0001	<.0001	<.0001	0.001	0.945
Non S&P Ret ₁	0.329	-0.037	0.098	0.057	-0.009	Non S&P Ret ₁	0.121	-0.100	0.104	0.035	-0.034
	<.0001	0.005	<.0001	<.0001	0.487		<.0001	<.0001	<.0001	0.005	0.005
	Non S&P 1986						Non S&P 1996				
S&P OF	0.665	0.097	0.262	0.006	-0.151	S&P OF	0.702	0.029	-0.043	-0.101	0.003
	<.0001	<.0001	<.0001	0.652	<.0001		<.0001	0.017	0.000	<.0001	0.797
N S&P OF	0.919	-0.008	0.113	0.027	0.033	N S&P OF	0.948	0.058	-0.018	0.017	0.001
	<.0001	0.526	<.0001	0.031	0.007		<.0001	<.0001	0.143	0.172	0.910
S&P Ret ₁	0.470	0.022	0.148	0.003	0.006	S&P Ret ₁	0.346	0.046	0.031	0.014	0.014
	<.0001	0.092	<.0001	0.833	0.628		<.0001	0.000	0.013	0.265	0.269
N S&P Ret ₁	0.515	-0.047	0.143	-0.010	0.041	N S&P Ret ₁	0.374	0.028	0.069	0.045	0.050
	<.0001	0.000	<.0001	0.430	0.002		<.0001	0.024	<.0001	0.000	<.0001

Table 8, Panel B: Principal Components of Returns

	PC1	PC2	PC3	PC4	PC5		PC1	PC2	PC3	PC4	PC5
	S&P 1986						S&P 1996				
S&P OF	0.821	0.233	0.043	0.017	-0.017	S&P OF	0.912	0.048	-0.033	-0.016	0.066
	<.0001	<.0001	0.000	0.172	0.172		<.0001	0.000	0.007	0.202	<.0001
Non S&P OF	0.538	0.328	0.042	0.013	0.048	Non S&P OF	0.668	0.045	0.018	0.222	0.117
	<.0001	<.0001	0.001	0.288	<.0001		<.0001	0.000	0.137	<.0001	<.0001
S&P Ret ₁	0.246	0.376	0.065	0.010	0.082	S&P Ret ₁	0.046	0.180	0.060	0.244	0.140
	<.0001	<.0001	<.0001	0.447	<.0001		0.000	<.0001	<.0001	<.0001	<.0001
Non S&P Ret ₁	0.183	0.281	0.057	0.018	0.049	Non S&P Ret ₁	0.039	0.173	0.079	0.227	0.129
	<.0001	<.0001	<.0001	0.169	0.000		0.001	<.0001	<.0001	<.0001	<.0001
	Non S&P 1986						Non S&P 1996				
S&P OF	0.657	0.234	0.035	-0.046	0.002	S&P OF	0.721	-0.092	-0.144	-0.025	-0.063
	<.0001	<.0001	0.004	0.000	0.891		<.0001	<.0001	<.0001	0.043	<.0001
N S&P OF	0.817	-0.049	0.034	0.033	-0.002	N S&P OF	0.848	0.051	0.055	-0.027	-0.017
	<.0001	<.0001	0.006	0.008	0.844		<.0001	<.0001	<.0001	0.026	0.162
S&P Ret ₁	0.489	0.018	0.068	0.049	0.023	S&P Ret ₁	0.279	0.026	0.162	-0.049	-0.047
	<.0001	0.178	<.0001	0.000	0.077		<.0001	0.033	<.0001	<.0001	0.000
N S&P Ret ₁	0.467	-0.059	0.067	0.117	0.038	N S&P Ret ₁	0.281	0.100	0.201	-0.042	-0.011
	<.0001	<.0001	<.0001	<.0001	0.004		<.0001	<.0001	<.0001	0.001	0.369